Compounding hazards increase flood economic losses across Europe

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1 Introduction and objectives

Compound events—combinations of multiple hazards that contribute to societal or environmental risk—can significantly exacerbate disaster impacts. However, their effect on flood-related losses remains insufficiently quantified. In this study, we compile a novel, pan-European multi-hazard dataset spanning 1981–2020 at sub-national resolution. By combining hazard information with regional exposure and vulnerability data, we apply an ensemble machine learning model with explainable AI to identify associations between compound hazard complexity and flood losses. This work builds on machine learning tools to inform policy on compound climate impacts. and offers insights that could support the development of multi-hazard early warning systems.

2 Methods and data

The analysis is based on the HANZE flood impact database (Paprotny *et al.* 2023). We combine it with hazard information from climate (Muñoz-Sabater *et al.* 2021) and hydrological (Tilloy *et al.* 2025) reanalysis to produce a pan-European compound hazard dataset spanning 1981–2020 at subnational resolution. We focus on five compound hazard event types: wet sequences, drought–flood, heatwave–flood, coldwave–flood, and windstorm–flood events. Following the typology of Zscheischler *et al.* (2020), we combine HANZE flood records with compound hazard information to generate an event-level compound hazard dataset. We examine yearly temporal trends in single floods versus compound hazard events across Europe using the Mann–Kendall test, and test whether compound hazard events are associated with systematically higher recorded losses than single floods. We then link all flood records from HANZE with local exposure, vulnerability, and compound hazard occurrences at NUTS3 scale. We compute the flood complexity index that summarizes the mix of compound hazard types within each region. This index, together with exposure, vulnerability, flood severity, is used in an ensemble machine learning framework to predict mean log-transformed flood losses. We complement this with a causal analysis using Double Machine Learning and a Causal Forest decomposition (Chernozhukov *et al.* 2018) to test whether hazard complexity is associated with increased losses after controlling for confounders.

3 Preliminary results

This dataset reveals that more than 70% of recorded flood events in Europe involve compounding hazards, with an increasing trend over time. Notably, the top 1% of these events, in terms of economic losses, are all compound hazard events. By applying machine learning approaches, including GBM with SHAP explanations and Causal Forests to estimate average treatment effects at the NUTS3 regional level, we show that compound hazard complexity, along with higher flood intensity and regional vulnerability, consistently contributes to elevated flood losses. Taken together, the temporal trends, loss distributions, and ML analyses provide convergent evidence that compound hazard complexity plays an

important role in shaping flood impacts. This underscores the importance of considering compound hazards in risk assessment and preparedness strategies.

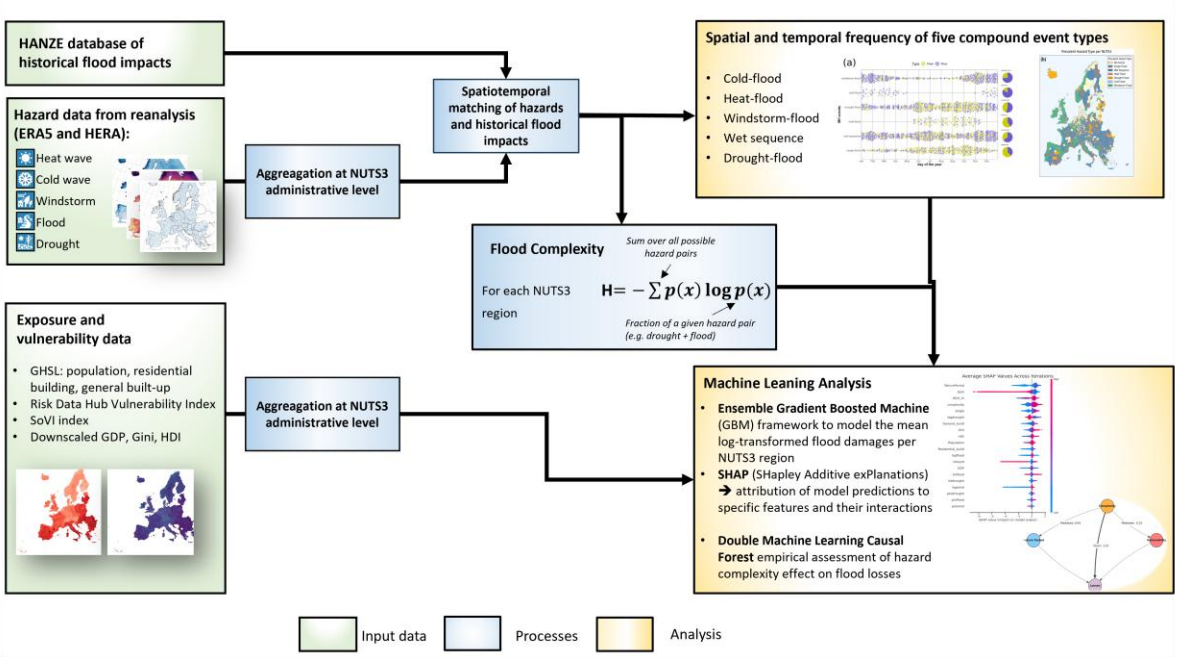

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Figure 1. Schematic summarizing the research.

References

Chernozhukov V, Chetverikov D, Demirer M, Duflo E, Hansen C, Newey W and Robins J (2018) Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal* **21**(1), C1–C68. https://doi.org/10.1111/ectj.12097.

Muñoz-Sabater J, Dutra E, Agustí-Panareda A, Albergel C, Arduini G, Balsamo G, Boussetta S, Choulga M, Harrigan S, Hersbach H, Martens B, Miralles D, Piles M, Rodríguez-Fernández N, Zsoter E, Buontempo C and Thépaut J-N (2021) ERA5-Land: A state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data Discussions* (January 1981), 1–50. https://doi.org/10.5194/essd-2021-82.

Paprotny D, Terefenko P and Śledziowski J (2023) An improved database of flood impacts in Europe, 1870 - 2020: HANZE v2.1. *Earth System Science Data Discussions* 1–37. https://doi.org/10.5194/essd-2023-321.

Tilloy A, Paprotny D, Grimaldi S, Gomes G, Bianchi A, Lange S, Beck H, Mazzetti C and Feyen L (2025) HERA: a high-resolution pan-European hydrological reanalysis (1951–2020). *Earth System Science Data* **17**(1), 293–316. https://doi.org/10.5194/essd-17-293-2025.

Zscheischler J, Martius O, Westra S, Bevacqua E, Raymond C, Horton RM, van den Hurk B, AghaKouchak A, Jézéquel A, Mahecha MD, Maraun D, Ramos AM, Ridder NN, Thiery W and Vignotto E (2020) A typology of compound weather and climate events. *Nature Reviews Earth & Environment* **1**(7), 333–347. https://doi.org/10.1038/s43017-020-0060-z.